

## Suggested FYP topics for 2026-27

by

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| Project code :             | YBL1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Project title :            | Vibe Networking – Network Engineering                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| No. of students :          | Up to 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Description of project :   | <p>This project explores the potential and capabilities of applying LLM to layer 2 (switching) and layer 3 (routing) network engineering. We are going to adopt Cisco's Modelling Lab (CML) network emulator which already has an open-source MCP server that can interface with LLM platforms. There are three goals: (i) assess the performance of existing LLMs on network configuration tasks by developing a LLM benchmark specifically for this use case; (ii) identify their limitations and develop solutions; (iii) explore LLM-driven automated network troubleshooting.</p> <p>Students who have taken IERG4090 or IERG4831 or are familiar with CML preferred.</p> |
| Industrial collaboration : | No.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

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| Project code :             | YBL2                                                                                                                                                                                                                                                                                                                                                                                |
| Project title :            | Vibe Networking – Network Protocol Research                                                                                                                                                                                                                                                                                                                                         |
| No. of students :          | Up to 2                                                                                                                                                                                                                                                                                                                                                                             |
| Description of project :   | <p>This project explores the application of LLM towards autonomous design, implementation, and evaluation of transport protocols. The LLM is to be equipped with the capability to conduct physical network emulation experiments to test the performance of transport protocols, collect trace data for analysis, and refine the protocol design iteratively and autonomously.</p> |
| Industrial collaboration : | No.                                                                                                                                                                                                                                                                                                                                                                                 |

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| Project code :             | YBL3                                                                                                                                                                                                                                                                                                                                                                                             |
| Project title :            | Vibe Networking – Network Traffic Analysis                                                                                                                                                                                                                                                                                                                                                       |
| No. of students :          | Up to 2                                                                                                                                                                                                                                                                                                                                                                                          |
| Description of project :   | <p>Real-world Internet services often generate massive amounts of logs and analytics. This project explores the application of LLM towards autonomous network traffic analysis. There are three goals: (i) to discover patterns that can be used for predictions; (ii) to detect anomalies; (iii) to discover correlations between performance and system/network configurations/topologies.</p> |
| Industrial collaboration : | No.                                                                                                                                                                                                                                                                                                                                                                                              |